

An agent-based model to quantify energetics, movement and habitat selection of mid-continent mallards in the Mississippi Alluvial Valley

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ABSTRACT

The dynamics of wintering waterfowl populations at the landscape scale are the result of complex interactions of environmental, behavioral and energetic drivers. Agent-based models provide a method to directly link these factors in a spatially explicit framework and allow the emergence of patterns from the aggregation of individual agent actions. We adapted the Spatially-explicit Waterbird Agent-based Model Program (SWAMP), originally developed for waterfowl in central California, to simulate a basin-scale population of mallards (*A. platyrhynchos*) wintering in the Mississippi Alluvial Valley over a four-month period (November–February). Simulated agents move within the landscape, foraging on areas made available based on a probabilistic inundation status, and converting food resources to endogenous energy. The model uses a high-resolution map of eastern Arkansas waterfowl habitats and incorporates a hierarchical habitat selection system that enables mallards to relocate at increasing scales in response to changing food availability. We validated the performance of modeled mallard body condition and behavioral metrics under a range of environmental conditions against expected outcomes derived from empirical data and found that the simulation produced realistic representations of changes in flight distances, energy expenditure, lipid storage, and foraging habitat use in response to depleting food resources over time. We discuss the model's applicability as a tool to quantify waterfowl response to a range of environmental conditions and to evaluate scenarios of landscape composition and configuration in the context of waterfowl population management.

1. Introduction

Many migratory waterfowl species rely on widely distributed habitats to meet seasonal life history requirements. Environmental conditions encountered at one stage of the annual cycle have the potential to influence body condition, survival, and breeding success in subsequent portions of the year (Dubovsky and Kaminski 1994; Devries et al., 2008; Sedinger and Alisauskas 2014). In North American waterfowl, cross-seasonal effects between winter and the following spring migration and breeding season are common; birds in better body condition at the end of winter achieve better demographic outcomes (Guillemain

et al., 2007; Sedinger and Alisauskas 2014). Therefore, the management of wetland landscapes on the wintering grounds is crucial for the conservation of North American waterfowl (Davis et al., 2014). Such management relies on understanding how waterfowl respond to site-specific habitat conditions (e.g., food availability) in the context of the rest of the landscape, and what bioenergetic benefits they derive (Gray et al., 2013; Kaminski and Elmberg, 2014). Empirical methods such as population surveys and telemetry can provide some pattern-based information on current habitat use versus availability, but they do not make process-based predictions of habitat use under other landscape scenarios. Body condition studies are primarily descriptive,

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and they do not mechanistically explain how changes in habitat quantity, quality, or distribution might affect bioenergetics. It is clearly not feasible to experimentally manipulate entire landscapes to empirically test habitat use and bioenergetic predictions, let alone contrast multiple scenarios. A spatially explicit model that monitors the effects of habitat quantity, quality, and spatial configuration on bird movements and bioenergetics could provide waterfowl managers with a tool to evaluate the effects of large-scale conservation and management actions on waterfowl populations.

Agent-based models (ABMs) provide a powerful approach to simulate the functioning of landscape-forager systems. The system's dynamics arise from the interaction of individual actors with each other and their shared environment, enabling the bottom-up emergence of large-scale patterns from small-scale interactions (Grimm and Railsback 2005; McLane et al., 2011). Wintering waterfowl populations are particularly well-suited to ABMs because foraging and movement costs differ widely between foraging locations and vary over time, the location of habitats relative to each other is important, and links between habitat conditions and metrics at both the individual and population level are of interest (Williams et al., 2014). Over an extended timeframe, wintering waterfowl use foraging and resting habitat at scales beyond the normal daily foraging range in response to changing food availability or seasonal behavioral shifts (Beatty et al., 2014; Johnson et al., 2014; Ringelman et al., 2015; Yetter et al., 2018); thus, habitat selection decisions are hierarchical. In addition, modeling at large scales allows consideration of spatial and temporal heterogeneity in the distribution of waterfowl and habitat types. However, such changes in spatial extent, landscape complexity, and number of simulated agents are likely to reveal issues of scale dependence that require explicit accommodation of inter-scale transitions (Fritsch et al., 2020). In ABMs applied at spatial and temporal scales appropriate to encapsulate behavior and distributions of wintering waterfowl populations, agents must be able to deal with complex landscape configurations that may pose behavioral and energetic challenges different from those encountered at smaller scales. To our knowledge, this challenge has not yet been addressed, yet it is a necessary step in the development of models useful in the conservation management of waterfowl populations at behaviorally realistic scales.

The Spatially explicit Waterbird Agent-based Model Program (SWAMP; Miller et al. (2014), Dugger et al. (2021), Eadie et al. (2023), *in prep.*) was developed as a multi-species ABM to simulate the actions of large numbers of waterbird agents on a foraging landscape of dynamically depleting food resources. The model was validated on small programmatic maps (100 km²) and applied to moderate-size GIS landscapes (~900 km²) in the Central Valley of California, where it was shown to accurately represent the expected energetics and movements of generalized waterfowl agents undertaking foraging flights at this scale (Miller et al., 2014; Eadie et al., 2023, *in prep.*). Building on this model foundation, we here describe the parameterization and application of an expanded version of SWAMP capable of simulating waterbird foraging on landscapes of increased scale and complexity. The model was adapted for a wintering mallard (*Anas platyrhynchos*) population on a ~29,000 km² simulated landscape in the eastern Arkansas region of the Mississippi Alluvial Valley (MAV). The landscape mosaic of the MAV at the scales of interest, containing major river courses as well as woody and herbaceous wetland complexes within a matrix of different crop fields, is of substantially greater complexity than the relatively homogeneous, agriculturally-dominated landscape of the Central Valley.

We selected mallards as the model species because approximately half of wintering waterfowl in the MAV consists of mallards (US Fish and Wildlife Service, 2016), making the species an appropriate choice for modeling habitat and movement dynamics. We selected eastern Arkansas because of the region's exceptional importance for wintering ducks (Soulliere et al., 2013; US Fish and Wildlife Service, 2016), the availability of local observational data sets for model calibration (Krementz et al., 2012; Beatty et al., 2014) and because the area contained landscape composition representative of the MAV, making the results

generalizable over a wide region. Because an intended application of the model is the evaluation of scenarios involving the establishment and management of wetland conservation easements administered by the U. S. Natural Resources Conservation Service (NRCS), we also prioritized a region containing a large number of conservation easements.

We evaluated model outputs compared to reasonable ecological, physiological and behavioral expectations based on observed ecological measurements. We used an approach similar to pattern-oriented modeling (POM) (Grimm et al., 2005; Grimm and Railsback 2012) by identifying desirable model behaviors on several levels, followed by evaluation of how well these criteria were achieved, and if necessary calibration of parameters. We set three broad targets:

- A) Model output should satisfy multiple basic validation criteria:
 1. Agents disperse to foraging patches
 2. Daily metabolic expenditure increases throughout the season
 3. Daily energy requirements are within the range observed from empirical data
 4. Days to deficit (time until current population can no longer cover energy requirements) at largest population size exceed the length of the remaining season
- B) Seasonal dynamics of key agent metrics (population size and emigration, forage flight distance, habitat-specific forage time proportion, lipid storage, and food energy demand, intake, and net gain) should align with realistic expectations and display internal consistency in their interactions.
- C) Global environmental changes imposed on the map should affect the simulated population in predictable ways. Resource-rich conditions should reduce energy expenditure, improve body condition, and increase population retention longer into the season, whereas scarcity conditions should make foraging more energetically costly, less rewarding and ultimately, result in greater emigration from the region.

We discuss the extent to which the model meets these expectations, and its suitability for use in predictive simulations.

2. Model description

Our purpose in developing this expanded version of SWAMP was to produce a tool for conservation planners and waterfowl managers to evaluate the potential effects of landscape changes on populations of waterfowl foragers during the non-breeding season. We generated estimates of body condition, emigration rates, time/activity budgets, and habitat use, and then estimated landscape energetic carrying capacity from these values. The model was intended to be applicable to real-world landscapes of a scale and complexity relevant to the observed movements and behavior of wintering waterfowl populations.

2.1. The SWAMP model

The following is a summary of the structure and functionality of the model. An Overview, Design concepts and Details (ODD; Grimm et al., 2020) description of the full model is available in Eadie et al. (2023), *in prep.* An ODD description of the hierarchical patch selection system that was added in the present model version (see below) is provided as a submodel ODD in the Supplementary Material (B).

The model consists of a modified version of SWAMP (Spatially explicit Waterbird Agent-based Model Program; Miller et al. (2014), Dugger et al. (2021), Eadie et al. (2023), *in prep.*) which is implemented in Java using MASON (Luke et al., 2005), a library for the development of agent-based, spatially-explicit models. The model simultaneously simulates the behavior of independent agents foraging on a landscape with dynamic food energy values and availability. It can use algorithmically generated landscapes or make use of imported GIS maps (as in the present study). The simulation landscape is made up of variably

sized land cover units that are based on contiguous map polygons characterized by a set of parameters, referred to as “patches”. Patches are distinguished by their cover type, which determines food biomass and energy content, and flooding status. The simulated ecosystem represents wetlands and agricultural lands during winter when these cover types are often flooded, and thus it is assumed that food biomass is depleted and not restocked during that time period. Initial food biomass is distributed to patches at model initialization (using published mean values and a gamma distribution). Depletion occurs via two processes: a constant decay rate applied to all food items, and removal of food biomass by foraging agents.

Agents follow a generalized behavior pattern considered representative of dabbling ducks during winter, made specific to mallards where possible. Simulated birds spend most of the day in a fixed number of roosting locations and disperse in the afternoon to forage in nearby wetlands and crop fields, returning to roost at the closest roosting location following acquisition of sufficient daily food resources. No explicit flocking behavior is implemented, although flocking-like behaviors emerge from the aggregate behavior of agents. When moving to a patch, agents move towards its geometric center at a common flight speed and upon arrival are placed at a random location within the patch. To limit range of displacement by this mechanism, large patches were subdivided during map processing (see Supplementary Material [A.4]). Agents cannot change their movement target during flight, but are able to immediately select a new target on arrival if the target patch has become unsuitable in the meantime.

The probability of an agent selecting a specific food patch is proportional to food density and flight distance. This behavior is implemented as a hierarchical habitat selection system that allows agents to relocate at increasing distance scales in response to decreasing local food resources. We delineated two broader landscape categories, including the foraging area (a small assemblage of local patches) and the sub-basin (a larger assemblage of foraging areas). Under normal circumstances, an agent chooses a target foraging area, then a target patch within the foraging area, for the day's first foraging flight. However, if body condition indicates that foraging has been insufficient recently, the agent will first relocate to a different sub-basin. The selection of patches, foraging areas, and sub-basins uses different weightings of the perceived importance of food density and flight distance. This system was found to be fundamental for the realistic simulation of agent behavior at the employed large scale of this study. For details of this system, see the submodel ODD in the Supplementary Material (B).

Competition is indirect (exploitative), in that depletion of food density lowers the probability that another forager will select the same patch, and patches have a maximum forager density of $1/\text{m}^2$. If food density falls below a specified giving-up threshold, the patch is no longer considered as a foraging location. Agents will switch to a new patch during a foraging bout if they can no longer derive a rate of energy gain greater than their foraging metabolic rate from the current patch.

Agents can only forage on patches currently marked as flooded, representing at least partial inundation. Although empirical evidence indicates that mallards forage in dry agricultural fields (Jorde et al., 1983; Calicutt et al., 2011), we restricted foraging to flooded patches because the majority of wintering mallard foraging in the MAV occurs in inundated habitat (Edwards et al., 2012; Herbert et al., 2021). Flooding status for a given model run is applied probabilistically at the start of the simulation based on each patch's flooding probability (derived from satellite data; Allen 2016), yielding different patterns of available foraging areas across replicate runs of the same map. Foraging activity is governed by a type II functional response modeled using Holling's disk equation (Holling 1959):

$$i = \frac{\alpha' R t}{1 + \alpha' R H_T} \quad (1)$$

where i is amount of food consumed, t is elapsed time since start of

foraging in present patch (s), α' is an attack constant, R is food density in the patch, and H_T is handling time for a unit of food. The attack constant represents search efficiency (m^2/s). Foragers thus acquire food at a rate proportional to food density, modified by search and handling efficiency. Food energy is processed using a two-stage metabolism model. Following consumption, food-derived energy is directly added to a non-lipid energy reserve, roughly equivalent to blood glucose. At the end of the next roosting period, all remaining non-lipid reserves are moved to the lipid store, which represents body fat, using a set conversion efficiency (the conversion involves a loss of energy). Metabolic demands are first supplied from non-lipid reserves and then from lipid reserves; if both are depleted, the agent dies. Metabolic activity costs are computed as multiples of a base resting metabolic rate, specified for foraging and non-foraging activities; the latter are treated as one time-weighted average of a number of activities such as sleep, courtship, preening, etc. We used a consistent time step of 15 min during which costs are evaluated and actions are carried out.

Agents in the model emigrate out of the landscape (leave the simulation) if their stored lipid reserves drop below a certain threshold. This threshold is defined as a fraction of the agent's maximum lipid stores and is checked before the start of each day's foraging excursion. Thus, emigration will generally preempt death by starvation except in rare cases generated by the model's timing schedule. The model has no representation of external sources of disturbance, thus all agent movements and exits from the simulation are driven by foraging dynamics. Mallard mortality from sources other than hunting is considered to be minimal in the MAV during the wintering period (Reinecke et al., 1987; Dugger et al., 1994; Link 2007). Mortality was therefore not included as a factor affecting the number of mallards leaving the simulation, and the model focused on emigration as the sole exit mechanism.

Supplementary Figure A.1 shows a flow diagram of decisions and actions taken by an agent during one time step.

2.2. Parameterization

2.2.1. Agent parameterization

Table 1 shows the parameters used in the present model to define the behavior of individual agents and lists the sources used. For details on implementation, see Miller et al. (2014) (parameters 1–16), and Supplementary Material (B) (parameters 17–21). Notes on some specific parameters are given below.

Giving-up density, or the food biomass per patch below which agents will no longer select a foraging patch, should ideally be specified per food or habitat type (Williams et al., 2014), however the model currently only supports one value per forager type. A weighted mean of published giving-up values per habitat type (woody wetland: $1 \text{ g}/\text{m}^2$; herbaceous wetland: $20 \text{ g}/\text{m}^2$; corn: $1.5 \text{ g}/\text{m}^2$; rice: $5 \text{ g}/\text{m}^2$; soybeans: $5 \text{ g}/\text{m}^2$) (Gray et al., 2013) was therefore computed based on the proportion of habitat types in the map and applied to all foraging patches.

A mean attack constant for mallards, representing search efficiency (m^2/s ; α' in Holling's disk equation), could not be sourced from the literature except for values derived from experimental settings with unrealistically high food density and/or for peck-feeding, rather than the filter-feeding employed by mallards in wetlands. We therefore used raw experimental data from a study of teal (*Anas crecca*) filter-feeding on low densities of rice (Arzel et al., 2007), provided by Céline Arzel (C. Arzel, unpublished data), to estimate an attack constant appropriate for a filter-feeding dabbling duck in a natural setting ($0.00127 \text{ m}^2/\text{s}$; see Table 1).

The selection weights for patches, foraging areas, and sub-basins governed how strongly an agent's choice of a destination patch was driven by flight distance to the destination versus food density at the destination (see Supplementary Material [B]). The sub-basin switching threshold was the percentage of maximum lipid storage at which the agent decided to move to another sub-basin (large-scale subdivision of the foraging landscape) because the current one was yielding too few

Table 1

Values and sources for parameters of agent behavior at model initialization.

Parameter	Initial value	Unit	Source	Comments
Base metabolic rate	0.00136	kcal/s	Miller and Eadie (2006)	
Metabolic rate multiplier: foraging	2.5		Wooley and Owen (1978)	
Metabolic rate multiplier: flying	12		Miller et al. (2014)	composite from several sources
Metabolic rate multiplier: non-foraging	2		Miller et al. (2014)	weighted mean of several activities
Flying speed	21.4	m/s	Bruderer and Boldt (2008)	
Daily food intake limit	137.5	g	unpublished data	empirically sourced from diet/condition experiments by Eadie et al. (UC Davis; unpublished data)
Maximum lipid reserves	3463.1	kcal		
Starting lipid reserves	1978.9	kcal		
Starting lipid reserves, SD	494.7			
Conversion factor non-lipid <-> lipid	0.537		Miller and Newton (1999)	
Giving-up food density	3.16	g/m ²	Gray et al., 2013	area-weighted mean of values for individual habitat types
Maximum forager density per patch	1	1/m ²	Miller et al. (2014)	field observations
Foraging start time	17:30		Miller et al. (2014), unpublished data	composite from several sources
Foraging end time	09:30		unpublished data	
Attack constant	0.00127	m ² /s	unpublished data	calculated from the raw experimental data (C. Arzel, unpublished data) used in Arzel et al. (2007)
Handling time	2	s	Miller et al. (2014)	composite from several sources
Patch selection weights distance: food density	1: 3		empirical estimation from simulation tests	configured to satisfy model validation requirements (see Results)
Foraging area selection weights distance: food density	1: 3			
Sub-basin selection weights distance: food density	0.5: 3			
Sub-basin switching threshold	50%			
Emigration trigger threshold	15%		Whyte and Bolen (1988)	estimated from reported lipid stores

food resources. These values were calibrated to align with the observed movement patterns of mallards in the region (Beatty et al., 2014). Briefly, Beatty et al. (2014) used satellite-tracked mallard hens to derive a frequency distribution of mallard movements over different distances that shows a distinct peak in the frequency of movements between ~250 m and 30 km (interpreted as local relocations during foraging), and a much reduced frequency of longer movements ≥ 30 km

(interpreted as long-distance relocations). To derive simulated flight distances for this comparison, movement records for all agents were extracted from the model at 6 h intervals (median reporting frequency of the empirical movement data in Beatty et al. (2014)). Using iterative adjustments of selection weights and distribution of roosting locations (see 2.2.2. Landscape Parameterization, below), we minimized the difference between simulated and recorded median flight distance to a ratio of 5.5 km (model) vs 3.1 km (field data). For details about the calibration process, see Supplementary Material (A.6).

The trigger threshold for emigration was the percentage of maximum lipid storage at which the agent emigrates out of the simulation landscape. The value was based on findings by Whyte and Bolen (1988) that even mallards in relatively poor body condition (~57 g of body fat) were still able to migrate for up to 500 km in response to worsening weather conditions. This metric corresponded to a stored energy value of 513 kcal in the model, thus the threshold for emigrating out of the simulation due to reduced body condition was set to 513 kcal / 3463.1 kcal (maximum lipid reserves) = ~15%.

Fixed foraging start and end times ensured that all agents used the same daily schedule and determined the maximum duration of the foraging bout (16 h), although foraging terminated earlier if the daily food intake limit was met.

2.2.2. Landscape parameterization

The model was developed to represent foraging and movement behavior during the winter period, and map scale was chosen to enable adequate simulation of this behavior. During winter, mallards use foraging habitat mostly within a radius of ~30 km around a roost (Beatty et al., 2014), suggesting a minimum simulation scale of ~60 × 60 km if only a single central roost was present (Beatty et al., 2017). A larger scale was required to allow mallards to move among multiple roosting locations. To retain a connection to local landscape configuration and functional foraging habitat units, we based scale selection on the classes of USGS hydrologic units (HUCs; Jones et al., 2022), which are determined by watershed boundaries. We selected the 6-digit HUC (HUC6; 'Basin' size) as the most suitable unit area. With a mean area of 27,000 km², it is 7.5 times the size of the minimum simulation scale.

We selected a location in the MAV to represent by considering criteria of representative habitat distribution, amount and distribution of conservation easements, large wintering mallard population, and availability of field data for model verification. A circle of the required size was placed at the chosen location and the circumference was approximated with watershed boundaries at the HUC10 level ('watershed' size) to create the outline of the map. Fig. 1 shows the location of the selected area in east-central Arkansas, with a midpoint at 91° 26' N, 34° 40' E and an area of 29,353 km². For further details on the location selection process, see Supplementary Material (A.2).

The map was parameterized using spatial and qualitative data from the following publicly available databases: Cropland Data Layer (release 2019, CDL; US Department of Agriculture, 2018); National Wetlands Inventory (version October 2019, NWI; US Fish and Wildlife Service, 2019); Southeast Floodplain Inundation Frequency (Allen 2016); and National Conservation Easement Database (release 2018; National Conservation Easement Database, 2018). Extracts from two non-public databases [National Easement Staging Tool (NEST; last updated 2017); Public Lands Water Management Unit Inventory, maintained by the Lower Mississippi Valley Joint Venture (LMJVJ; winter 2013–2014)] were provided by the owners on request. For further details on the use and sourcing of these data, see Supplementary Material (A.3).

The vector-based map used a spatial resolution based on the main underlying data source (CDL; 30 × 30 m). Several passes of smoothing, merging, and sub-partitioning operations at the level of habitat patch units were carried out to simplify the map and reduce computational cost. For further details on these operations, see Supplementary Material (A.4).



Fig. 1. Location of selected simulation area in east-central Arkansas. Circle (red): area used for selection, with midpoint at 91° 26' N, 34° 40' E, size 27,000 km². Outline (black): area approximated using hydrologic unit (HUC) HUC10 boundaries, size 29,353 km². State lines are shown in gray and core Mississippi Flyway area is shown in black.

Parameterized foraging habitat made up 33.2% (9732 km²) of the total map area and consisted of five principal land cover categories that were used by agents: woody wetlands, herbaceous wetlands, corn, rice, and soybeans. Permanent open water was not used for foraging or roosting by mallards in the simulation. Wetland classification as implemented in the CDL and NWI databases follows the categories published by the [Federal Geographic Data Committee \(FGDC 2013\)](#), but categories were further combined and simplified for the simulation. Woody wetlands (50.9% of habitat) represented broad-leaved deciduous forested wetland with at least some amount of red oak species (*Quercus* spp.). Herbaceous wetlands (1.7% of habitat) represented wetland habitat containing non-woody vegetation. Soybeans, rice, and corn (respectively 25.6%, 20.2%, and 1.6% of habitat [47.4% total]) represented harvested fields of sown agricultural crops that are foraged for waste grain by wintering waterfowl. Woody and herbaceous wetlands were further distinguished by whether they were managed for waterfowl food biomass. For woody wetlands, this was connected to approximate percentage of red oaks in the forest composition. In order of increasing food biomass, classes were unmanaged (20% red oak; 97% of woody wetlands), managed (40% red oak; 3%), or oak-rich (70% red oak; <0.1%). For herbaceous wetlands, classes in order of increasing food biomass were restored (restored from former crop lands; 58% of herbaceous wetlands), unmanaged (18%), and managed (24%). Food biomass and energy content per mass unit for each patch were calculated as the sum of plant and invertebrate biomass, based on literature values. At start of simulation, each patch was assigned a food value sampled from a gamma distribution around this biomass mean. For further details on habitat categories and food parameters, see Supplementary Material (A.5). [Fig. 2](#) shows a map of the parameterized model world.

A binary flooded/non-flooded status determined whether a given patch was available for foraging during a simulation run. Categories of foraging habitats were assigned different flooding probabilities. At the start of the simulation, flooding status was probabilistically determined for each patch using its flooding probability and remained in effect throughout the run. This simplification of a constant patch inundation status was adopted to enable the generation of a different probabilistic

mosaic of flooded patches at the beginning of each replicate run. We defined four global flooding regimes to represent different map-wide environmental conditions, implemented by modifying all patch flooding probabilities by a common rule. These regimes were standard (~30% of total foraging habitat inundated/available; 2947 km²), flood (~54%; 5234 km²), weak drought (27%; 2640 km²), and strong drought conditions (~12%; 1129 km²). For further details on flooding probabilities and environmental conditions, see Supplementary Material (A.5.4).

After foraging, mallards returned to roost at the nearest of a limited number of pre-defined roost patches. These were forageable herbaceous or woody wetland patches with 100% flooding probability that were marked as roosting locations. We used 225 roosts, the number and distribution of which were determined in the process of calibrating the model to approximate the desired median flight distance (see 2.2.1. Agent Parameterization, above). This process was also used to determine the partitioning of the map for the hierarchical patch selection process. For further details about the calibration process, see Supplementary Material (A.6).

2.2.3. Simulation setup

For each simulation replicate we re-randomized patch inundation conditioned on patch-level flooding probability, as well as the distribution of food resources around a fixed mean. This process ensured that every replicate run featured a different mosaic of exploitable foraging patches on the landscape. We conducted an *a priori* power analysis to determine the number of replicates required to provide a representative range of outcomes, and found that 136 replicate runs per scenario would satisfy our criteria (for further details, see Supplementary Material [A.7]).

We used a simulation duration of 120 days, representing the period from November 1 to March 2, as covering most of the wintering period in Arkansas. We used a nominal starting population size of 212,500 mallards, which represents the median of Arkansas Mid-Winter Count numbers for the period 1981–2016 ([US Fish and Wildlife Service, 2016](#)), scaled to simulation area. To ensure a realistic rate of food resource exploitation, we included an equal number of “non-mallard” dabbling ducks because the ratio of mallards to all other waterfowl in the region is roughly 1:1 ([US Fish and Wildlife Service, 2016](#)). In the absence of species-specific parameterization, they were parameterized as mallards in all respects, but were not evaluated in outcome metrics, with the exception of some metrics directly connected with total landscape energy (total demand, total intake, and days to deficit). Rather than initiating the model at full starting population size, we added groups of immigrating agents over the period of day 1–30 (November 1 – November 30) until the full amount was reached, basing the daily proportion of arrivals on a sigmoid distribution modeled on empirical data ([Krementz et al., 2012](#); [Beatty et al., 2014](#)) (Supplementary Fig. A.5). Arriving agents were initially distributed among roosting locations proportional to roost area. For presentation of results, all data series were left-truncated on day 30, excising the first month of simulation because the repeated addition of new individuals during that time largely obscured dynamics. All inference was thus based on the 90-day period between day 30 and day 120 only.

Model runtime was reduced by implementing a scaling method that allowed reducing computation load without sacrificing data accuracy or resolution. Briefly, the initial number of agents placed on the map, area of each patch, and food biomass assigned to each patch were divided by a scaling factor. Agent interaction with patches used the scaled area, but movement distances were based on unadjusted values, ensuring that energetic expenditures and travel times were preserved while energetic values of patches were adjusted accurately for a smaller population. Tests with a scaling factor of 1/100 showed that scaling did not affect simulation outcomes, but greatly decreased simulation time. This scaling was thus used for all simulations.

Model world

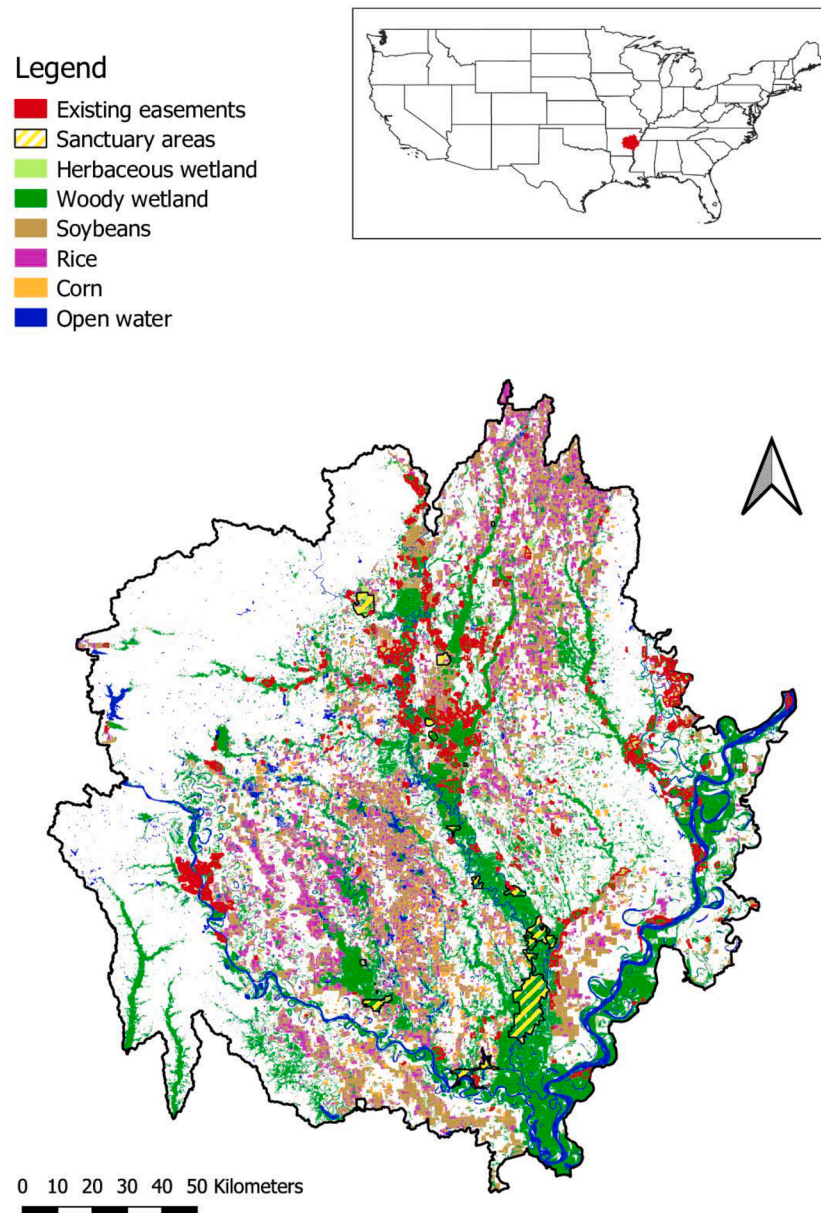


Fig. 2. Habitat type map of modeled area. All areas assumed not to be used by foraging mallards, other than open water, have been removed (non-parameterized; white). The parameterized area represents 33.2% of the total area (9732 km²). Easement (red) and sanctuary areas (yellow) are distinguished by outlines and barred fills. Inset shows location within continental United States.

2.3. Code and model validation

Core model functionality is based on the previously validated SWAMP model (Miller et al., 2014). An in-depth code review was conducted for all code, and versioning was used to verify that each new function or submodel performed as intended. Proposed use-cases were tested with a wide range of parameters at high replication prior to generating outcomes for formal evaluation.

Several criteria were evaluated to validate basic model functionality. These criteria were identical in formulation to the validation targets set for the previous model version in Miller et al. (2014) but targeted to values specific to the present model and simulated region.

- 1 *Agents disperse to foraging patches* in dynamic response to changes in food availability and foraging efficiency (qualitative). This criterion

was considered met if mallard's foraging time budgets for the model's five principal foraging habitat types (herbaceous and woody wetlands, soybeans, rice, and corn) showed clear seasonal changes.

- 2 *Daily metabolic expenditure increases throughout the season* (qualitative). This criterion was considered met if mallards consistently incurred increasing energy costs with progressing time of year.
- 3 *Daily energy requirements are within the range observed from empirical data* (quantitative). The estimated daily energy requirement of a "generalized dabbling duck" is 294.5 kcal/day (Gray et al., 2013); Miller et al. (2014) used a target range of 225–280 kcal/day for a combination of waterfowl species. Because mallard body mass tends to be toward the upper end of the range of body mass for dabbling ducks, we considered this criterion to be met if mean agent energy demand over the season fell within a range of 240–295 kcal/day.

4 Days to deficit at largest population size exceed the length of the remaining season (quantitative). Days to deficit is the number of days until the average forager could no longer meet current daily energy requirements, given current available landscape food energy. The modeled landscape was expected to provide at minimum sufficient energy to sustain the population for the duration of the modeled season. The number of days to deficit changes each step as landscape energy and the demands on foraging behavior change, and in procedural models is thus of limited use as a forecasting metric; for greatest utility, it may best be evaluated at the time of maximum population size. Therefore, we considered this criterion met if median days to deficit on day 30 (just after immigration into the region was completed) was > 90 days (length of remaining season) for the complete model population, including non-mallard foragers.

3. Results

3.1. Basic model validation

All criteria for the validation of basic model functionality were met by outputs under standard environmental conditions.

- 1 *Mallards dispersed to foraging patches* in dynamic response to changes in food availability and foraging efficiency, resulting in nonlinear seasonal patterns of foraging behavior as food resources depleted at different rates (see Supplementary Figs. C.1–5). See below (Section 3.2) for details on exploitation patterns.
- 2 *Metabolic costs increased throughout the season* after an initial period of settling into optimal early exploitation locations, and mean demand (energy expenditure that must be satisfied by acquiring food energy) increased correspondingly and consistently thereafter (see Supplementary Figures C.6 and C.7).

3 *Daily energy requirements were within the range observed from empirical data.* Mean energy demand over the season showed median values of 255–270 kcal/day, with a bilateral 95% value range interval of 245–285 kcal/day (see Supplementary Figure C.7), which was within the target range of 240–295 kcal/day.

4 *Days to deficit at largest population size exceeded the length of the remaining season,* being well in excess (490 days) of the required threshold (90 days) at the point of greatest population size (day 30), and remaining above it throughout the season (see Supplementary Fig. C.8).

3.2. Dynamics of output metrics and response to environmental conditions

We evaluated whether the temporal dynamics of agent output metrics, and their response to increased or decreased food availability, corresponded to realistic physiological and behavioral processes and showed internal consistency. We examined the output of the following metrics: population size, emigration percentage, forage flight distance, forage time proportion per habitat, energy demand, energy intake, net energy gain, and lipid storage. Results are shown as medians for the four environmental conditions (standard [green], flood [blue], weak drought [brown], strong drought [yellow]), for days 30–120 of a 120 day run-time over 136 replicates. The emergent character of the simulation produced substantial possible outcome spaces across replicates; the boundaries of these are shown as bilateral 95% value ranges (shaded areas).

3.2.1. Population size and emigration

Population size of mallards present on the map constantly declined under all conditions due to emigration of agents from the simulation (Fig. 3A). The decline was slower under flood than standard conditions, slightly faster under weak drought than standard conditions, and much more rapid under strong drought, where numbers frequently reached

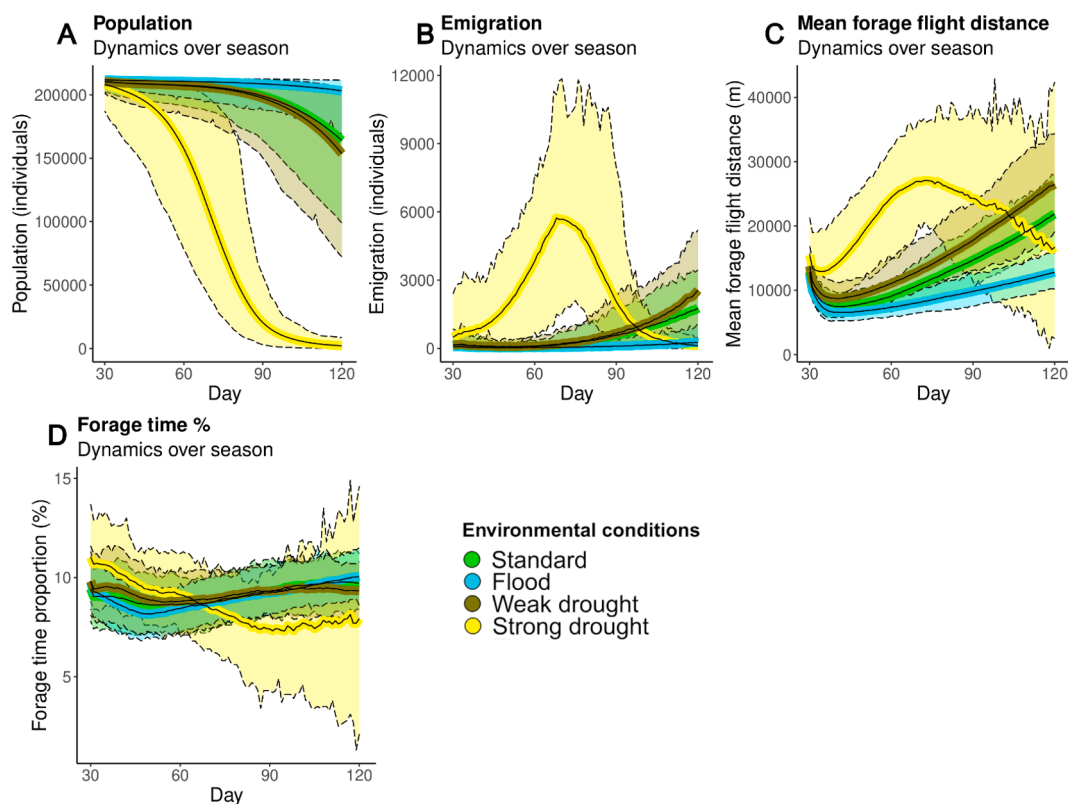


Fig. 3. Dynamics of model metrics for mallards over the season for the four environmental conditions. Medians with bilateral 95% value ranges (shaded areas) are shown. A, population size; B, daily emigration; C, mean first forage flight distance of the day; D, mean percentage of daily time budget spent foraging.

0 (entire population emigrated). Mallards emigrated (left the simulation) when their stored energy dropped below the emigration threshold. Cumulative emigration numbers were inversely correlated to population size. Under most conditions, daily emigration continually increased over the season as food availability declined; emigration was slower in flood and more rapid in weak drought conditions, while under strong drought the rapidly dropping population led to a mid-season peak in daily emigration followed by a decline (Fig. 3B).

3.2.2. Forage flight distance

Forage flight distance was the mean distance that a mallard had to travel from the roosting site to the first foraging patch of the day (Fig. 3C). This movement and the return flight to the roost following sufficient food consumption comprised the large majority of daily movements in the simulation, and therefore represents a useful indicator of food accessibility at any point. Flight distances were at a minimum early in the season when mallards had settled into optimal early exploitation positions, then increased when food resources in patches close to roosts became depleted and patches with high food energy became more sparse on the map. Flood conditions required shorter and drought conditions longer flight distances than standard conditions. Under strong drought conditions, flight distances peaked mid-season and then declined when the population had dwindled to a few individuals exploiting the best remaining patches.

3.2.3. Foraging time

The mean proportion of a day mallards spent foraging fluctuated around 9% (133 min) throughout the season, increasing towards the end in response to depleting food resources (Fig. 3D). The main limiting factor on foraging time was the daily intake mass limit (137.5 g) which could be reached in much less time than was available for the whole foraging bout (16 h), even under strong drought conditions. Consequently, 80–90% of each day was spent in roosting behavior.

The modeled habitat types were present in different proportions and had different food biomasses per unit area, energy values, and flooding probabilities (Supplementary Tables A.1–3), resulting in specific exploitation patterns. Flooded crop patches (47.4% of forageable area) had less food biomass than wetlands ($\sim 7.5 \text{ g/m}^2$ vs $33\text{--}78 \text{ g/m}^2$) but contained food of greater relative energy value ($\sim 3.4 \text{ kcal/g}$ vs $\sim 2.5 \text{ kcal/g}$) and were thus prioritized early by mallards (Fig. 4A). Herbaceous wetlands were rare in the landscape (1.7% of foraging area) and had lower food energy value than crops, but had much greater potential food biomass and were relatively likely to be flooded under all

conditions (Supplementary Tables A.1–3), and thus were slow to deplete as a category. Proportional use increased early in the season as soybeans were exhausted (see below), then leveled out to $\sim 25\%$ (Fig. 4B). While woody wetlands had lower food energy and biomass than crops, and lower flooding probabilities than the other foraging categories (Supplementary Tables A.1–3), they made up $\sim 50\%$ of the foraging area and were associated with roost-rich sanctuary areas (Fig. 2). They therefore contributed 30–50% of exploited habitat under all conditions (Fig. 4C). Their proportional use increased until mid-season under most conditions as the food biomass in crop patches was depleted, then leveled out or declined slightly as low-yield patches became depleted and mallards started prioritizing still-productive rice fields or herbaceous wetlands. Under strong drought, foraging focused on this most common remaining habitat type.

Soybeans were the most common crop but had the lowest biomass and most rapid decay (Supplementary Table A.2), and therefore were depleted first (Fig. 5A). Rice had greater biomass per unit area and was more likely to be flooded under all conditions than other crops (Supplementary Table A.3), providing foraging throughout the season. Rice use therefore increased in the late season as food biomass in some wetlands became depleted (Fig. 5B), and the dynamics observed in the general crops category tended to be driven by rice. Corn patches were rare (1.6% of forageable area), less likely to be flooded than other crop types, and contained moderately large biomass; they remained available at low but constant exploitation volume throughout the season (Fig. 5C). Proportional crop use declined early in the season under all but strong drought conditions as food biomass on soybean patches was reduced and ducks shifted to wetland patches, then increased again when food biomass in wetlands thinned out and mallards made more use of the reliable late-season rice availability (Fig. 4A). Under strong drought, the high flooding probability of rice patches made them more attractive and more readily exploited than other crops (Fig. 5B). All crop food biomass was subject to more rapid decomposition (over the course of the season, 22.6% for rice, 42.7% for corn, and 88.9% for soybeans) than food resources in wetlands (8.3% for woody and 19.4% for herbaceous wetlands; Supplementary Table A.2), which contributed to the early-season shift to wetland foraging. Late-season crop (rice) use was lower under flood conditions because more wetland resources remained available.

3.2.4. Energy demand, intake, and lipid stores

Energy demand was principally driven by distance and frequency of foraging flights, and energy dynamics therefore closely followed mean foraging flight distance (Fig. 4D, 6A). Mean per capita energy demand

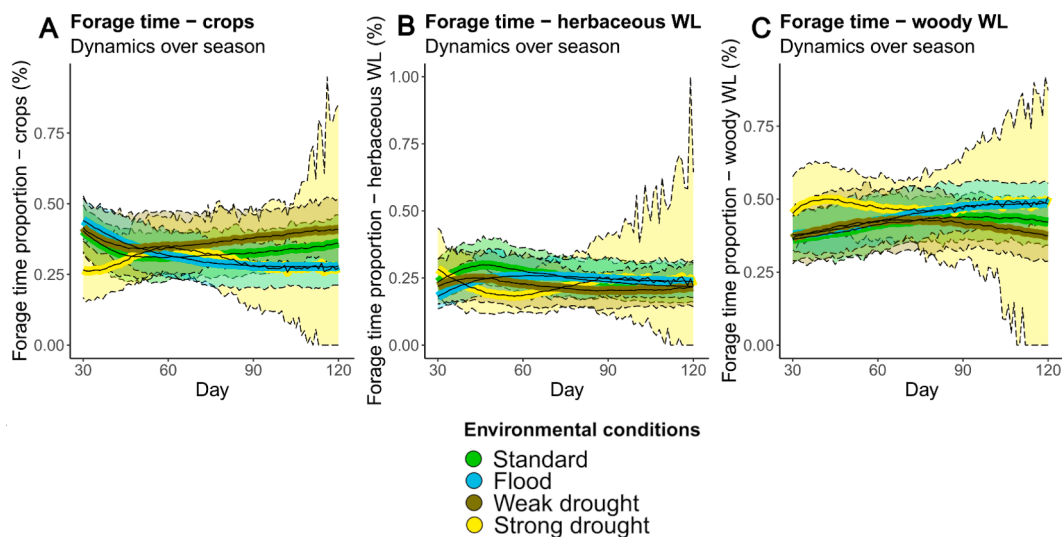


Fig. 4. Dynamics of mean percentage of daily time budget that mallards spent foraging in major habitat types over the season, for the four environmental conditions. Medians with bilateral 95% value ranges (shaded areas) are shown. WL: wetlands. A, crop patches of any type; B, herbaceous wetlands; C, woody wetlands.

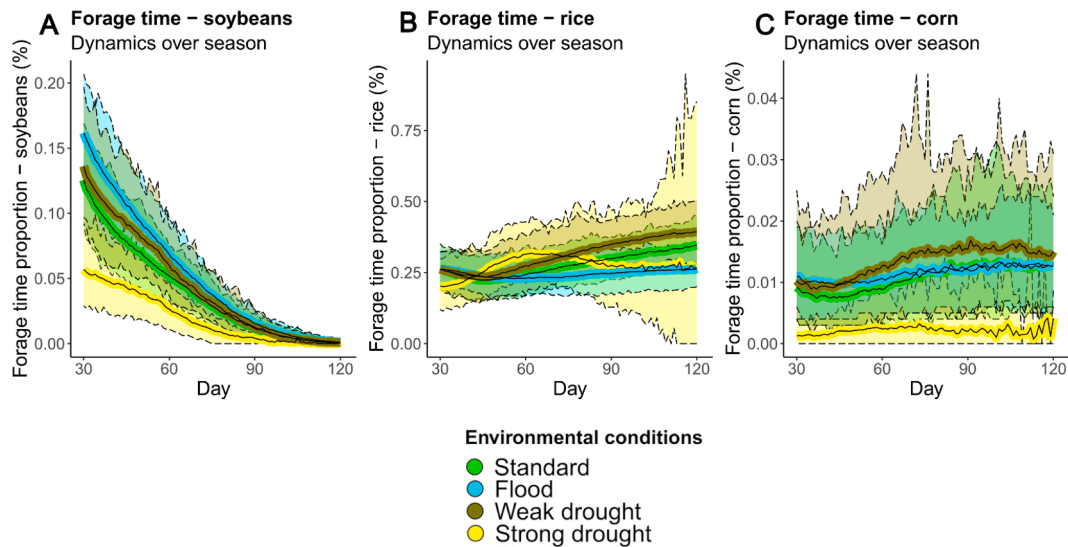


Fig. 5. Dynamics of mean percentage of daily time budget that mallards spent foraging in three main types of agricultural (crop) patch over the season, for the four environmental conditions. Medians with bilateral 95% value ranges (shaded areas) are shown. A, soybeans; B, rice; C, corn.

scaled with increasing food scarcity and continually increased over the season as patches with food resources became more widely distributed; demand was lower under flood and higher under weak drought conditions. Under strong drought, mean individual demand declined in the later season because the few remaining individuals became more stationary in the remaining productive food patches with little competition and thus lower depletion rates.

Energy intake was capped in the short term by an individual's daily mass intake limit (137.5 g, irrespective of a food item's energy content) and in the longer term by its lipid storage limit (3463 kcal). Intake was at maximum at the start of the season (Fig. 6B) as agents foraged in a

resource-rich landscape, achieving large net energy gains (Fig. 6C) and consequently filling up their lipid storage (Fig. 6D). At the end of this period (~day 50), the maximum lipid storage was reached under most environmental conditions and intake leveled out. As the season progressed, this level was maintained, despite progressive landscape food depletion, by self-censoring of the population – individuals with continually negative net energy gain eventually depleted their lipid stores sufficiently to emigrate, ensuring that the remaining population continued to be dominated by individuals with intakes above this energy level. Mean intake continued to follow the rise in demand throughout the remainder of winter, with some deficits incurred under weak

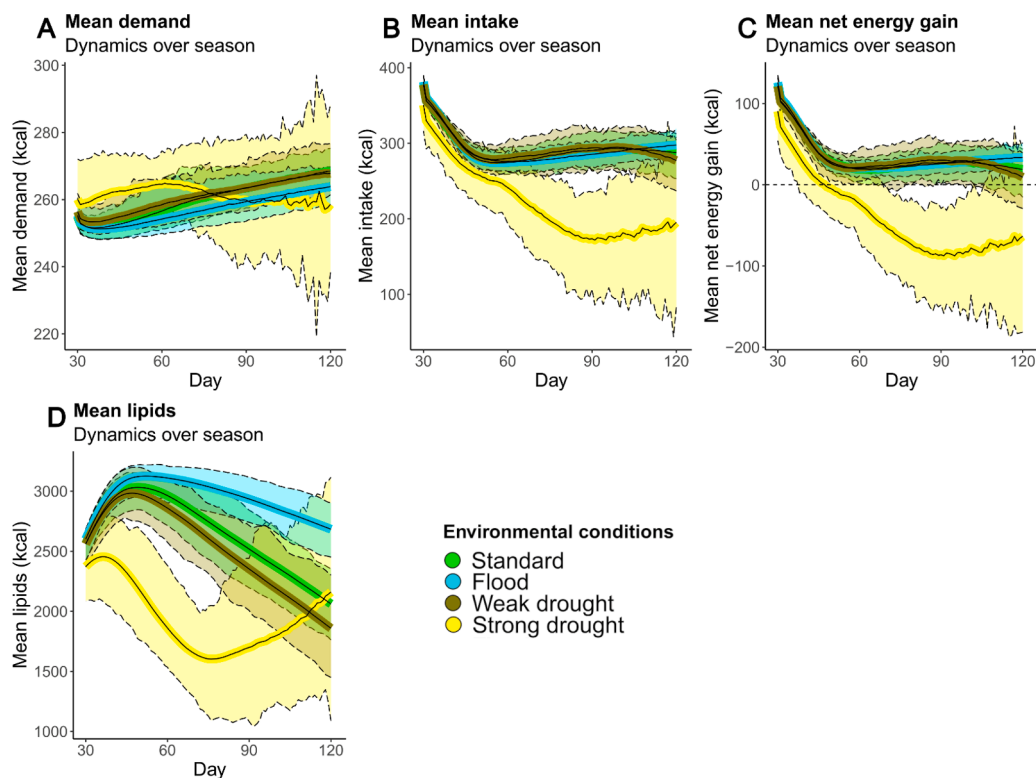


Fig. 6. Dynamics of model energy metrics for mallards over the season for the four environmental conditions. Medians with bilateral 95% value ranges (shaded areas) are shown. A, mean daily energy demand incurred; B, mean daily energy intake; D, mean daily net energy gain; E, mean daily lipid stores.

drought conditions at end of the season (Fig. 6A, B). However, lipid stores of all individuals continually declined from this point onward due to a combination of two causes (Fig. 6C). First, a depleting landscape offered progressively fewer opportunities to meet the daily intake limit with food of the highest energy content (principally rice). Second, the daily scheduling in the model imposed a guaranteed lipid loss at the beginning of each day because remaining metabolic energy was converted to lipids (at a loss) just before the start of the first foraging flight, which thus had to be “paid” for from lipid stores.

Under strong drought conditions, the majority of mallards could not forage efficiently enough after the early season to maintain a positive energy balance (Fig. 6C), let alone meet the lipid storage limit (Fig. 6D), and intake and energy gain decreased continually. The late season increase in mean net energy gains and mean lipid stores under these conditions was driven by individuals with low energetic resources emigrating from the simulation, and the remaining individuals enjoying reduced competition.

4. Discussion

The expanded SWAMP model performed well in simulating the wintering population of mallards in a large-scale region of the MAV. The strengths of ABMs in simulating ecological systems include the spatially-explicit representation of habitat data and agent behavior; the provision of explicit inferences about metrics such as habitat selection and movement patterns at the individual level; the integration of individual actions at the population level, allowing the simulation of emergent effects at multiple scales; and the high-resolution tracking over time of multiple parameters (Grimm and Railsback 2005). The outputs of the present model illustrate these characteristics, documenting the individual and aggregate dynamics of a large population of mallard agents interacting with a dynamic landscape.

Agents' landscape-wide knowledge of food availability likely does not represent the true underlying process for foraging patch selection in mallards, but we specifically used this approach because it successfully re-created observed patterns with limited computation time. The behavior and physiological metrics of agents generally reflected real-life expectations, indicating that landscape and agent parameterization were suitable to represent the landscape-forager system. The four basic model validation criteria were equivalent to those investigated for the simpler previous version using programmatic landscapes (Miller et al., 2014). These verified that the use of GIS landscapes at a substantially larger scale (29,000 km² vs. 900 km²), combined with a behavioral model that allowed agents to efficiently forage at such scales, preserved the fundamental functions of the base model. We found that these criteria were consistently satisfied.

4.1. Seasonal dynamics

Agent metrics exhibited seasonal dynamics throughout the simulated period that both showed sensible causal relationships and were consistent with empirical data. Depletion of food patches, mediated by shifting availability and relative pay-off of different forage types, led to extended flight distances and consequently increased energy expenditure, which in turn was reflected in lower overall net energy gain and depleted body fat stores, ultimately driving the proportion of agents emigrating out of the simulation. While these metrics and their interplay follow reasonable expectations for a population of foragers in a landscape being depleted of resources, the simulated process of accruing lipid stores was subject to some limitations. The model does not include energetic demands on agents other than those arising from the changing patterns of food distribution on the map. In reality, energetic demands and body fat reserves of wintering waterfowl fluctuate in response to weather conditions, disturbance, and other influences that were not represented in the simulation (Jorde et al., 1984; Whyte and Bolen 1984; Morton et al., 1989). The model also assumes an invariant lipid metabolism with a

single conversion efficiency, whereas a migratory dabbling duck's metabolism actually would begin accumulating extra lipid stores towards the end of the wintering season in preparation for spring migration (Arzel et al., 2006; Newton 2020). Finally, the guaranteed lipid cost of the foraging bout's first daily flight imposed an artificial drain on lipid stores throughout the simulated period. The net effect of the latter two factors would be a more rapid decline of lipid stores in the second half of the season. However, the main driver of this decline was the progressive scarcity of high-energy forage patches close to roosting locations, forcing longer foraging flights (see Matchett et al. (2021) for observational evidence of this dynamic).

Under most environmental conditions, foraging habitat use in the model was characterized by consistent substantial use of herbaceous wetlands, strong use of woody wetlands increasing until mid-late season, a rapid depletion of soybeans, and consistent use of rice with increasing importance in the late season. The relatively slow decomposition rate of rice compared to soybeans and corn, together with high flooding probability of this patch type, incentivized increasing rice patch use towards late season in the simulation. This pattern agrees with observations that abundant food sources with a high decomposition rate, such as some crops, are exploited by waterfowl early in the winter (Herbert et al., 2018), while longer-lasting sources, like natural wetlands, become important later (Arzel et al., 2006); but remaining cultivated crops are attractive targets for food generalists like mallards even later in the winter (Pearse et al., 2012; Beatty et al., 2014). Mosaics of crops and natural wetlands may thus constitute a useful landscape pattern for wintering ducks (Davis et al., 2014). While use of woody wetlands was high throughout the simulated period, the model lacked a representation of disturbance avoidance, which would be expected to further shift the focus of foraging activity to forested wetlands that are designated as wildlife sanctuaries or are close to such areas (Arzel et al., 2006; McDuie et al., 2021).

4.2. Response to environmental conditions

The model was developed with the capacity to represent foraging landscapes under inundation regimes deviating from long-term average conditions. While such “wet” or “dry” winters in the MAV would normally feature varying frequencies of locally restricted flooding events of some days duration (Edwards et al., 2012), these could not be implemented in the current model system. We instead used simplified full-season modifications of flooding probabilities that globally either increased (flood) or decreased (weak and strong drought) flooded habitat availability, with a wide range of landscape mosaic outcomes. The generation rules employed (see Supplementary Material [A.5.4]) created differences in inundated habitat area between flood and strong drought conditions of $\sim 4.5 : 1$ (Table 2). This pattern is comparable to the lower limit of Reinecke et al. (1988)'s estimate of 5–10-fold differences in potential waterfowl habitat along major river courses in Mississippi and Arkansas during dry and wet winters. The short-term effects of flooding events on mallard physiology and behavior may be substantial; e.g., birds may engage in hyperphagia in response to flood events, increasing their consumption limits and daily foraging effort and dispersing to newly available forested habitats offering oak mast and invertebrates (Heitmeyer 2006).

However, few data are available about the expected reaction of duck populations to season- and landscape-scale flooding condition changes. Herbert et al. (2018) found that the extent and distribution of surface water was the best predictor of presence and abundance of wintering mallards in Arkansas, and Davis et al. (2011) suggested that in years with lower inundation conditions, emigration rates were greater for mallards wintering in the lower MAV. Veon et al. (2023) reported that mallard body mass in the lower MAV generally increased with seasonal precipitation and river flooding. Body condition and survival of northern pintails wintering in Texas were above average in wet years (Smith and Sheeley 1993) and below average in dry years (Moon and Haukois

Table 2

Flooded foraging area (km^2), available food energy from flooded area at simulation start ($\times 10^9$ kcal), and population size at simulation end, for the four implemented environmental conditions. Magnitude relative to the standard condition value is also shown for each metric, and magnitude relative to nominal starting population (212,500 mallards) for end-of-season population size.

Environmental conditions	Flooded foraging area (km^2) (% of Standard)	Available food energy ($\times 10^9$ kcal) (% of Standard)	Population size at end of season (% of Standard) (% of starting population)
Standard	2946.98 (100%)	61.76 (100%)	171,100 (100%) (81% of starting)
Flood	5234.13 (178%)	106.43 (172%)	206,900 (121%) (97% of starting)
Weak drought	2640.54 (89%)	55.53 (90%)	159,450 (93%) (75% of starting)
Strong drought	1128.95 (38%)	22.47 (36%)	1250 (1%) (1% of starting)

2006). Consequently, we expected the model to reproduce these broad qualitative shifts. In simulations under standard conditions, ~20% of mallards emigrated by day 120. Under natural high flooding conditions, the population exhibited the expected lower flight distances and energy expenditures, and consequently better lipid store build-up and population retention (~3% emigration). Weak drought conditions had the effect of requiring greater energy expenditure, resulting in increased emigration (~25%) and lower lipid stores. Strong drought conditions were severe by design, to illustrate possible responses to rare and extreme unfavorable conditions. Under strong drought, rapid population decline occurred in days 60–90 (January) and reduced the population to close to zero at end of winter (~99% emigration). Mallard abundance data from the U.S. Fish and Wildlife Service (USFWS) midwinter count in several Arkansas regions show differences of up to 95% over a period of 35 years (US Fish and Wildlife Service, 2016), suggesting that population fluctuations of this magnitude do occur. To calibrate landscape-scale models of waterfowl populations, future surveys that connect changes in inundation patterns with changes in waterfowl abundance and distribution at a greater temporal resolution than such single seasonal snapshots would be highly valuable.

Population responses to different inundation conditions demonstrate the emergent nature of higher-level model outputs. Relative changes in flooded foraging area translated essentially linearly into changes in available food energy but diverged for population retention (Table 2). Even considering that population size was bounded between zero and starting population and thus would be expected to follow an asymptotic distribution close to extremes, the response to strong drought conditions in particular was clearly greater than additive (a loss of 62% of flooded habitat resulting in the loss of 99% of the population, compared to –11% resulting in –7% under weak drought). Most of the more derived metrics, e.g., foraging time budgets (Figs. 4, 5), showed similar non-linear and emergent dynamics. This pattern is indicative of the capacity of models that simulate individual behavior to represent complex aggregate outcomes at the population level.

4.3. Applications and further development

The expanded SWAMP model has a wide range of applications in the exploration of landscape-forager interactions, conservation planning, and landscape management. This type of wildlife ABM can be regarded as a synthesis of dynamic optimization model, movement model, and land-use model (McLane et al., 2011), and as such provides a number of advantages over the bioenergetic models (such as “duck energy-day” [DED] models) that are widely employed in conservation management to develop estimates of landscape carrying capacities and bird population responses to management actions. Although they require greater implementation time and data, models such as SWAMP enable the spatially explicit tracking of metrics over time and the observation of

interactions of different levels of the simulated system as they play out. For example, temporal dynamics may inform decisions about when specific management actions are best implemented, based on changing body conditions or factors such as overlap in food sources exploited by different species. The modeled landscape in SWAMP can also dynamically represent environmentally-driven changes. While not used in the application presented here, the model is able to implement scheduled alterations in availability of different food sources throughout the year, for example due to sowing and harvesting operations or from irrigation and other water management. This is an important capacity when the simulation is intended to cover multiple months or is even cross-seasonal.

The complex nature and emergent behavior of large-scale ABMs should motivate the researcher to explore the possible outcome space, allowing the identification of the most likely trajectories of the metrics of interest (Marceau 2008). In addition, the stochastic nature of ABMs can reveal statistically unlikely but potentially important outcomes of scenario sets that could help identify critical habitat or environmental thresholds, coupled with spatial and temporal information. For example, such scenario planning can highlight areas that are more prone to unlikely events, and others that may remain strongholds of bird use in even the least likely scenario configurations. In our scenarios, the most extreme conditions simulated with the present model (strong drought) generally created the widest range of possible outcomes (Figs. 3–6), highlighting worst-case situations under which the survival of the remaining population depended on foraging patches with specific characteristics (high inundation probability, deep food biomass reservoirs, and in close proximity to roosting areas). The SWAMP model's flexible interface between model engine and GIS source data lends itself well to scenario planning using a library of maps representing possible future landscape compositions, management approaches, or environmental regimes. For example, we conducted a preliminary evaluation of the contribution of restored wetland conservation easements to wintering mallard populations; the effects of the extent and spatial distribution of newly established easements; and future conservation scenarios that may be most likely to benefit wintering waterfowl populations.

The current limitations of the model fall into two categories: those that may convey extra benefits to agents (such as absence of hunting disturbance and mortality, and absence of severe weather) and those that may impose extra costs on agents (such as absence of seasonal behavior and metabolism changes, and daily lipid cost). Accordingly, it is difficult to determine whether the population dynamics generated here are over- or underestimated, as no applicable full-season population data are available. However, two simplifications in the model that would have exacerbated agents' perception of “scarcity” beyond realistic values should be noted: first, the averaged giving-up food density used (3.16 g/m^2) was greater than the value specific to woody wetlands (1 g/m^2), curtailing exploitation of this most common habitat type earlier than strictly indicated. Second, agents made no use of un-flooded habitat (particularly dry crops), reducing the potential foraging landscape a mallard could use (Jorde et al., 1983; Calicutt et al., 2011). Concurrently, open water bodies, some of which might be used for foraging in harsh winter conditions (Heitmeyer 2006; Beatty et al., 2014; Davis et al., 2014), were excluded due to the difficulty of distinguishing them from unsuitable areas (e.g. flowing water). Given these constraints, we suspect that conditions of food availability in the model were overall more limiting than might actually occur in real-world situations.

Several substantial gaps in current knowledge about habitat use by waterfowl at the landscape scale had to be addressed by approximation in setting up this model. These included the scales at which mallards might make movement decisions based on food availability, and the number of distinct roosting locations that a mallard population might use over the wintering period. The approximation of flight distances observed in the field was used to estimate parameter values for these

processes (see Supplementary Material [A.6]). Results indicated that the model may not be capable of entirely matching observed flight distances through modification of these agent and landscape parameters alone. The proportional lack of short movements that would correspond to within-patch relocation in the model is a limitation of the fundamental model architecture, and probably causally connected to the larger median distance (5.5 km vs 3.1 km) we found in simulated between-patch flights, since agents could not move across patch boundaries in a slow and energy-efficient manner (walk or swim). This will have resulted in increased energy expenditure, with accumulating effects throughout the simulation. Individual tracking studies with high temporal resolution, which have become increasingly feasible and affordable due to advances in transponder technology and might form an efficient integrated complement to the high-resolution population counts suggested above, might help fill this data gap.

We found that minor variations in the availability of types of foraging patches could have major effects on agent energetics (see also Miller et al., 2014), suggesting that, for example, a longer-lasting availability of food from woody wetland patches might have a noticeable effect on emigration rates in the last month of the simulation (Fig. 3B). On the other hand, periods of harsh winter weather can have severe effects on mallard body condition (Jorde et al., 1984; Whyte and Bolen 1984), and hunting pressure strongly influences behavioral patterns of ducks, leading to avoidance of heavily hunted areas as well as prioritization of safe foraging and roosting locations (Dooley et al., 2010a; 2010b; Lancaster et al., 2015; McDuie et al., 2021), which may result in greater competition and more rapid food depletion on patches that meet these conditions. In further development, a detailed representation of hunting pressure and mortality, in tandem with a behavioral system that allows agents to incorporate perceived levels of disturbance into their selection of roosting and foraging locations, will be an effective next step in enhancing the model's reliability and representativeness. This will also allow a detailed investigation of the interplay between the distribution of roosts in the landscape and the dynamics of the modeled population. We also plan to implement a climate/weather system to drive habitat conditions on a map, influence behavioral patterns, and trigger migration decisions between multiple maps.

5. Conclusions

The expanded SWAMP model performed well in producing behavioral, movement and energetic outcomes that are comparable to empirical data and exhibit realistic seasonal dynamics. Although the model currently incorporates only landscape-energetic drivers, we suggest it is suitably descriptive and internally consistent for use in the comparative analysis of scenarios that predict the response of waterfowl populations to variations in landscape composition, habitat quality, and food availability. Our findings demonstrate that many population metrics are dependent on the interaction of dynamic and spatially-explicit parameters. Critically, static estimates of energetic carrying capacity as would be generated by simpler bioenergetic models would substantially overestimate the ability of this landscape to support wintering waterfowl. The expanded SWAMP model can be used to identify the principal drivers of waterfowl habitat use and bioenergetic conditions, and to test possible actions intended to manage these. While our applications so far have been focused on mallards, the model can readily be adapted to any waterfowl species, or assemblage of species, for which sufficient parameterization data are available, and the hierarchical selection framework of agent movement allows flexibility in the choice of simulated landscape scale. The future incorporation of climatic and disturbance factors should further enhance model ability to represent complex spatio-temporal population dynamics.

CRedit authorship contribution statement

Florian G. Weller: Conceptualization, Methodology, Software,

Validation, Writing – original draft, Writing – review & editing, Visualization. **Elisabeth B. Webb:** Conceptualization, Methodology, Writing – review & editing, Supervision, Project administration, Funding acquisition. **Sean Fogenburg:** Conceptualization, Methodology, Software, Validation, Writing – original draft, Writing – review & editing. **William S. Beatty:** Conceptualization, Methodology, Investigation, Writing – review & editing. **Dylan Kesler:** Conceptualization, Methodology, Writing – review & editing. **Robert H. Blenk:** Conceptualization, Methodology, Software, Validation, Writing – review & editing. **Kevin M. Ringelman:** Conceptualization, Methodology, Writing – review & editing. **Matt L. Miller:** Conceptualization, Methodology, Software. **Céline Arzel:** Investigation, Writing – review & editing. **John M. Eadie:** Conceptualization, Methodology, Investigation, Writing – review & editing, Supervision, Project administration, Funding acquisition.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.ecolmodel.2023.110488](https://doi.org/10.1016/j.ecolmodel.2023.110488).

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